CUDA SFM LM progress

Timing breakdown and a general-solver baseline

Dataset: Dubrovnik 135 | dense Schur | 3 LM iterations

0.426 s
Current graph API
mean of 3 timed runs

0.138 s
CUDA backend
specialized dense-Schur path

Hybrid
Baseline question
CPU nonlinear work + GPU solve

This update has two parts: where the current API path spends time, then whether a general CPU/GPU separation is plausible.

1. **Profile current path**
   Measured
2. **Separate general baseline**
   Proposed
3. **Measure full LM convergence**
   Next

Current implementation

Measured

The public graph API adapts generic GTSAM objects into a specialized CUDA SFM backend.

NonlinearFactorGraph + Values

Generic graph, current variable state, and factor ownership.

Specialized graph conversion

Extract SFM factors, keys, measurements, cameras, and points into host arrays.

CUDA SFM arrays / backend

Pack, upload, run LM with dense Schur, and retain packed result arrays.

Downloaded Values

Rebuild optimized camera and point Values from host-side result arrays.

CPU merge

Merge SFM results into the original Values and update optimizer state.

Boundary responsibilities

Boundary	What it does	Why it matters
Generic to SFM	CPU representation conversion	The CUDA backend does not consume arbitrary GTSAM factors.
SFM to device	Array packing and backend setup	Includes host builds as well as GPU work.
Result to Values	CPU reconstruction and merge	Object/state handling is included in end-to-end time.

Framing: this profile measures an integration path, not only GPU kernel time.

Whole graph API: 0.426 s

Measured

Measured scope: warm-up excluded, mean of 3 runs.

426.223 ms

Graph API total

optimizer + optimize + result/error

The API denominator includes construction and final result/error queries, not just optimize().

- Optimizer construction 36.203 ms (8.49%)
- optimize() 376.192 ms (88.26%)
- Result/error queries 13.828 ms (3.24%)

Component	Time	% of graph API
Optimizer construction	36.203 ms	8.49%
optimize()	376.192 ms	88.26%
Result/error queries	13.828 ms	3.24%

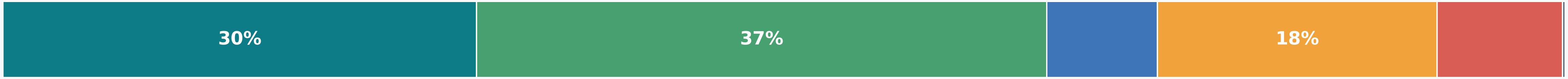
Scope reminder

The following slides unpack the 376.192 ms optimize() call.

Inside optimize(): 0.376 s

Measured

Measured components. Stacked bar: % of optimize(); table: % of Graph API.



- Graph conversion 114.246 ms
- CUDA backend 137.542 ms
- Backend return / assignment 26.685 ms
- Values merge / state update 67.464 ms
- Converted-data destruction 30.247 ms
- Residual 0.008 ms

Measured component	Time	% of whole API
Graph conversion	114.246 ms	26.80%
CUDA backend	137.542 ms	32.27%
Backend return / assignment	26.685 ms	6.26%
Values merge / state update	67.464 ms	15.83%
Converted-data destruction	30.247 ms	7.10%
Residual	0.008 ms	0.00%

CPU result management

97.711 ms

Values merge plus temporary converted-data destruction

Graph conversion and CPU result-management are prominent costs beside the backend.

Inside the CUDA backend: 0.138 s

Measured

Measured hierarchy within the specialized backend.

Backend layer	Time	% of backend
Setup	67.544 ms	49.11%
Solve loop	29.968 ms	21.79%
Download Values	40.030 ms	29.10%

Setup: projection host build 38.686 ms

The largest setup detail is CPU construction of the projection batch.

Solve loop: 29.968 ms

Dense Schur 23.009 ms | Hessian / damping diagonal 2.979 ms | linearized error change 3.655 ms

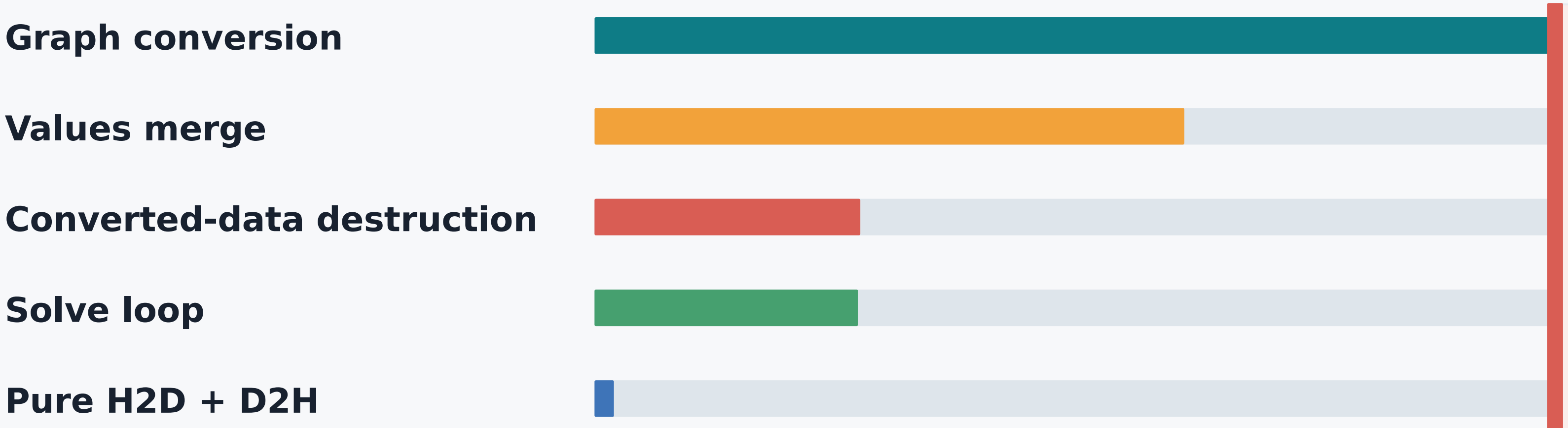
Download Values: 40.030 ms

Raw D2H 0.615 ms | Values rebuild 15.537 ms | remaining host allocation, wrapper, destruction, and tail 23.878 ms

What the profile says

Measured

Measured specialized path: representation conversion and state management dominate the current bottleneck.



Measured component	Time	% of graph API
Graph conversion	114.246 ms	26.80%
Values merge	67.464 ms	15.83%
Converted-data destruction	30.247 ms	7.10%
Solve loop	29.968 ms	7.03%
Pure H2D + D2H	1.936 ms	0.45%

Core conclusion

The present bottleneck is CPU-side representation conversion and object/state management, not PCIe copies or the dense-Schur solve.

This conclusion applies to the specialized representation measured here.

General-solver baseline

Proposed

Proposed Ceres-style separation, not an exact description of every Ceres path.

CPU / TBB factor evaluation + linearization

Arbitrary GTSAM factors stay on CPU.

❯

Reusable sparse numeric J / H buffers

Cache structure; refresh numeric values.

❯

H2D

Upload generic linear-system numeric data.

❯

GPU linear solve

Normal equations plus cuDSS or dense solver.

❯

D2H delta

Return only the solved update.

❯

CPU Values retract + trial decision

Manifold retract, error, and trust-region decision.

Why this is general

Arbitrary GTSAM factors remain in the CPU nonlinear layer. The GPU receives a generic linear system rather than an SFM-specific graph representation.

What remains a hypothesis

Sparse packing, repeated numeric upload, delta mapping, and end-to-end convergence must be measured in the actual prototype.

Sources: [Ceres installation] | [Ceres nonlinear least-squares]

Baseline feasibility timing

Measured | Estimated

Separately timed, synthetic per-candidate lower-bound arithmetic. Repeat-0-excluded sensitivity view.

Measured CPU operation	Mean	Role
CPU / TBB linearize	28.316 ms	Factor evaluation and linearization
Values retract	14.332 ms	Apply a prebuilt compatible delta
Trial graph.error	11.440 ms	Evaluate trial state
CPU subtotal	54.088 ms	Illustrative arithmetic, separately timed

Synthetic GPU reference

8.051 ms

Hessian diagonal + dense Schur combined - projection

62.139 ms

Synthetic lower bound

54.088 ms CPU + 8.051 ms GPU

+11.283 ms

Estimated generic transfer

115.1 MB / 9.50 GiB/s

73.422 ms

Lower bound + estimate

still not end-to-end

Caveat: not end-to-end. Sparse packing/layout, repeated upload, delta mapping, and convergence remain unmeasured. Do not compare these per-candidate synthetic figures directly to the 426.223 ms three-iteration API total.

Current assessment and next work

Assessment | Proposed

Measured evidence justifies a hybrid baseline experiment, not a performance claim.

Current assessment

Separately timed CPU nonlinear subtotal is tens of milliseconds per candidate versus the prior 0.426 s three-iteration API run. Scopes differ, so this supports feasibility, not a speedup claim. Expected benefit: generality and removal of specialized graph conversion.

Measured conclusion

Next work

- **Cache sparse structure.**
- **Direct TBB numeric fill.**
- **Upload only numeric values.**
- **GPU normal equations + cuDSS/dense solver.**
- **Download delta, then CPU retract/error.**
- **Measure full LM convergence.**

Success criterion: a measured end-to-end LM comparison with the same problem, convergence policy, and accounting scope.

Appendix: full dense-Schur breakdown

Measured

Measured mean, Dubrovnik 135, warm-up excluded, 3 runs. Times in milliseconds.

Level	Component	Time	% graph API	% parent
Graph API	Optimizer construction	36.203	8.49%	-
Graph API	optimize()	376.192	88.26%	-
Graph API	Result/error queries	13.828	3.24%	-
optimize	Graph conversion	114.246	26.80%	30.37%
optimize	CUDA backend	137.542	32.27%	36.56%
optimize	Backend return / assignment	26.685	6.26%	7.09%
optimize	Values merge / state update	67.464	15.83%	17.93%
optimize	Converted-data destruction	30.247	7.10%	8.04%
backend	Setup	67.544	15.85%	49.11%
backend	Solve loop	29.968	7.03%	21.79%
backend	Download Values	40.030	9.39%	29.10%

Detail	Time	Detail	Time
Setup: projection host build	38.686 ms	Solve: dense Schur	23.009 ms
Setup: pack values	13.688 ms	Solve: damping diagonal	2.979 ms
Setup: allocate trial values	11.769 ms	Solve: linearized error change	3.655 ms
Transfer: total H2D memcpy	1.526 ms	Transfer: total D2H memcpy	0.615 ms
Download: Values rebuild	15.537 ms	Pure H2D + D2H	2.141 ms

Note: the 1.936 ms specialized pure-copy figure cited in the feasibility summary comes from a separate transfer-focused run; this appendix uses the dense-Schur breakdown above.

Appendix: feasibility ranges and caveats

Measured | Estimated

Measured operations are separate loops. Derived hybrid values are sensitivity views, not sequential candidate calls.

Operation	All 30 mean [min, max] / CV	Repeat-0-excluded 27 mean [min, max] / CV
CPU / TBB linearize	30.851 [25.119, 57.299] / 26.39%	28.316 [25.119, 38.918] / 9.73%
Values retract	14.260 [12.320, 18.152] / 8.79%	14.332 [12.591, 18.152] / 8.71%
Trial graph.error	11.644 [8.518, 16.235] / 20.00%	11.440 [8.518, 14.845] / 19.26%
GPU projection	0.384 [0.379, 0.391] / 0.79%	0.384 [0.379, 0.391] / 0.82%
GPU Hessian diagonal	0.919 [0.909, 0.924] / 0.39%	0.919 [0.909, 0.924] / 0.41%
GPU dense Schur combined	7.513 [7.468, 7.576] / 0.42%	7.516 [7.468, 7.576] / 0.43%

First-index sensitivity

Index-aligned CPU subtotal: repeat 0 mean 80.752 ms, repeat 1 mean 56.881 ms, repeat 2 mean 57.024 ms. Repeat 0 is materially slower.

Exact transfer formula and provenance

553,336 x (24 Jacobian + 2 residual) x 8 B = 115,093,888 B = 115.1 MB. / 9.50 GiB/s = 11.283 ms. Dataset: dubrovnik-135-90642-pre.txt; AMD EPYC Milan, NVIDIA A100 80 GB PCIe; CUDA, driver, and TBB thread configuration not recorded.

Scope caveats: CPU loops and GPU references are independently timed; the synthetic arithmetic does not measure sparse packing/layout, repeated upload, delta download/mapping, or LM convergence. All-sample lower bound: 64.803 ms. Repeat-0-excluded lower bound: 62.139 ms. With estimated transfer: 73.422 ms.